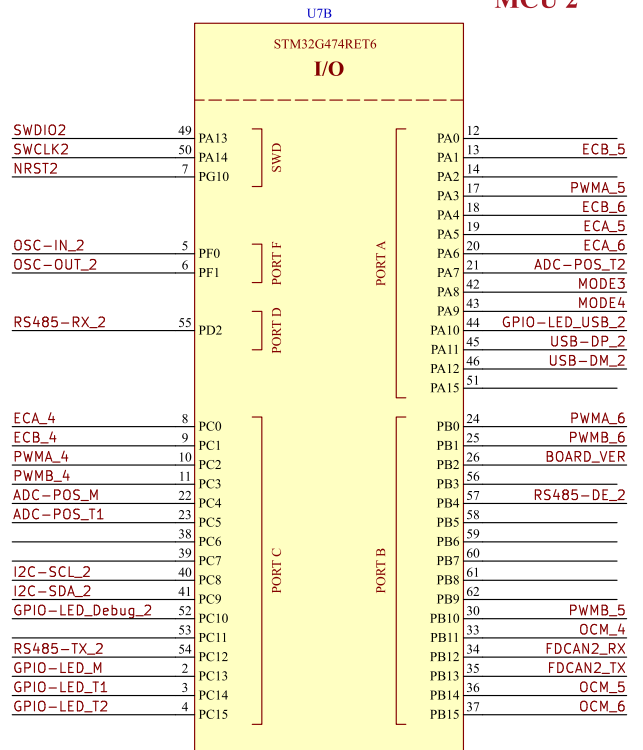
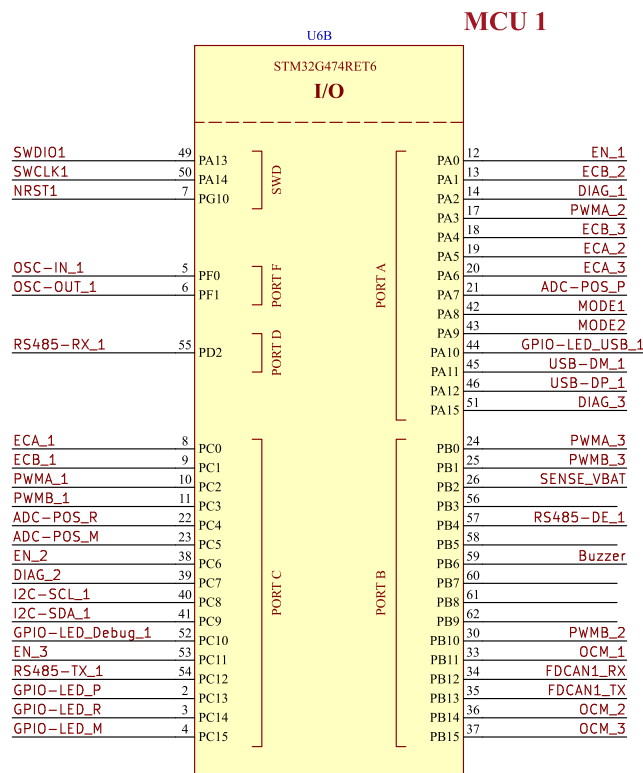


[5] MCU - IOs



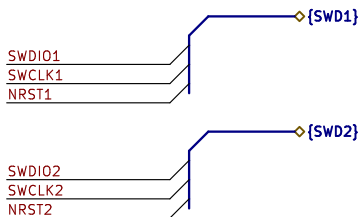
SENSE_VBAT SENSE_VBAT

Battery Voltage Sense

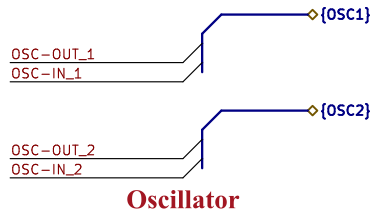
BOARD_VER BOARD_VERSION

Board Version Selection

Auxiliary Ports



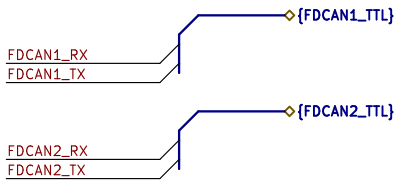
SWD (Debug) Port



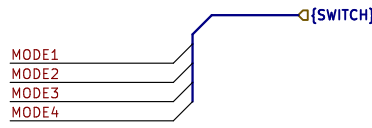
Oscillator

Buzzer Buzzer

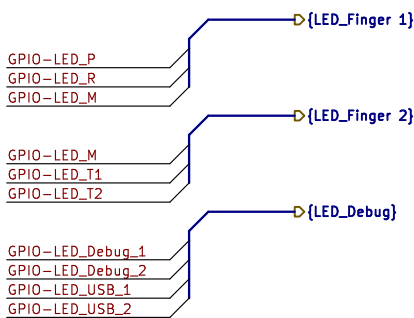
Buzzer



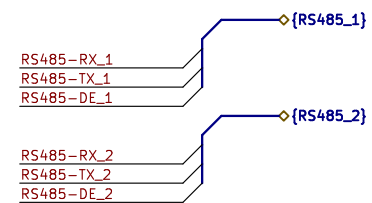
5Mbps FD-CAN Bus



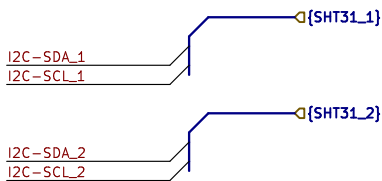
Switch



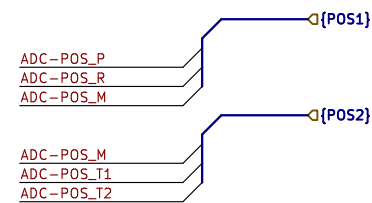
LEDs



RS - 485 Port



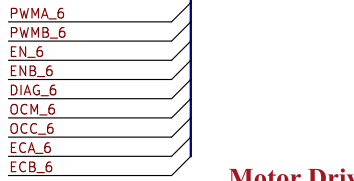
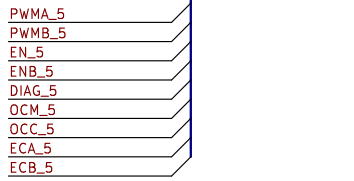
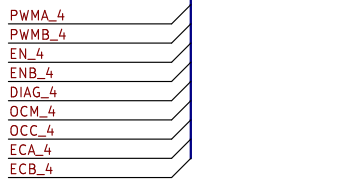
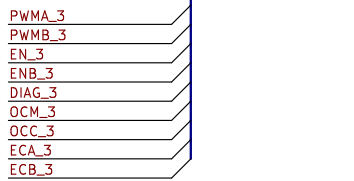
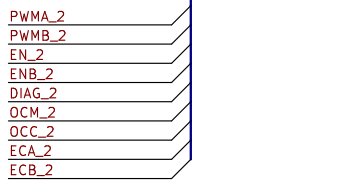
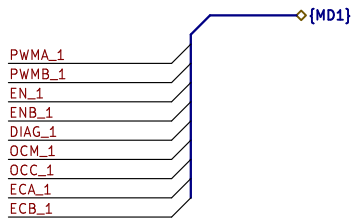
Temperature and Humidity



Hall Sensor - Position



USB PORT



Motor Driver

Motor Driver

PWMA_1 TP1
PWMB_1 TP2
DIAG_1 TP3
OCM_1 TP4

PWMA_2 TP5
PWMB_2 TP6
DIAG_2 TP7
OCM_2 TP8

PWMA_3 TP9
PWMB_3 TP10
DIAG_3 TP11
OCM_3 TP12

PWMA_4 TP13
PWMB_4 TP14
DIAG_4 TP15
OCM_4 TP16

PWMA_5 TP17
PWMB_5 TP18
DIAG_5 TP19
OCM_5 TP20

PWMA_6 TP21
PWMB_6 TP22
DIAG_6 TP23
OCM_6 TP24

Debug

SWDIO1 TP25
SWCLK1 TP26
NRST1 TP27

SWDIO2 TP28
SWCLK2 TP29
NRST2 TP30

FD-CAN

FDCAN1_RX TP31
FDCAN1_TX TP32
FDCAN2_RX TP33
FDCAN2_TX TP34

RS-485

RS485-RX_1 TP35
RS485-TX_1 TP36
RS485-DE_1 TP37

RS485-RX_2 TP38
RS485-TX_2 TP39
RS485-DE_2 TP40

Temperature & Humidity

I2C-SDA_1 TP41
I2C-SCL_1 TP42
I2C-SDA_2 TP43
I2C-SCL_2 TP44

Hall Position

ADC-POS_P TP45
ADC-POS_R TP46
ADC-POS_M TP47
ADC-POS_T1 TP48
ADC-POS_T2 TP49
ADC-POS_T3 TP50

Test Points

DESIGN NOTE:
We use the LQFP64
version to have extra
pins for controlling an
external fan.

DESIGN NOTE:
Do not forget to flash
option bytes when
flashing for the first
time.

LAYOUT NOTE:
Footprint deviates from
datasheet and IPC7351 to
allow for soldermask sliver
between pads according to
PCBWay design rules.

Comments:	Company: WILLAS 威倫電子股份有限公司 Willas Electronic Corp.	Variant:	Git Hash:
Sheet Title: MCU - IOs	Board Name: MOTOR CONTROL BOARD	Project Name: MASTER HAND	
Sheet Path: /Project Architecture/MCU - IOs/	File Name: MCU - IOs.kicad_sch	Designer: DAMON	Date: 2026-06-27
		Reviewer:	Size: A3
			Sheet: 5 of 25